

Arnav Karnik

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EXPERIENCE

Data Science Analyst | Capital One | Payment Pricing & Portfolio Analytics

Sep 2025 - Jun 2026

- Built a full-stack dynamic pricing simulation platform leveraging **Streamlit** as the frontend and **Snowflake** as the backend, to model and analyze how adjusting key pricing metrics **impacts payment network revenue**, enabling data-driven pricing strategy decisions
- Added industry-category attributes to the snowflake base table and surfaced the feature to the Streamlit UI, enabling **higher-level simulations** across grouped merchants and merchant category codes.
- Implemented a merchant profile module allowing simulations to apply custom weights across different segments

AARC Data Science Analyst | Discover | Finance Python Automation

Aug 2024 - Sep 2025

- Executed Current Expected Credit Loss (**CECL**) forecasting models across all of Discover's business units, aligning model outputs aligned with expected performance metrics, **improving forecast accuracy by 23%**.
- Built an interactive **Streamlit** app to validate **CECL** Card input driver data quality, visualizing over **1M+ records** monthly and enabling **30% faster identification of anomalies** through time series and null analysis comparisons against historical benchmarks.

Data and Analytics Intern | Discover Financial Services | Credit BI

Jun 2023 - Aug 2023

- Spearheaded the design and development of interactive dashboards using **Figma** and **Tableau**, leading to a **40% increase in user engagement** and improved decision-making efficiency for business partners.
- Collaborated with cross-functional teams to gather requirements, ensuring the **dashboards** meet stakeholder's needs and facilitated **data-driven decision-making**.

Associate Software Developer | Sentry

May 2022 - May 2023

- Utilized the **MVC** framework in **ASP.NET Core** to design and implement a link farm feature, allowing users to dynamically manage over **25+ data resource links**, resulting in a **30% reduction in support requests** related to outdated or missing references.
- **Built a robust API endpoint to reprocess failed file ingestions**, recovering and reloading over **1,000+ failed files** into **Sentry's data lake**, improving data completeness and reducing manual intervention by **75%**.

PROJECTS

Free Throw Percentage Tracker | ReactJS, Firebase, TailwindCSS, Python

- Designed and implemented a dynamic free throw tracking platform using **ReactJS** and **MaterialUI** for the frontend, with **Firebase** powering the backend, enabling users to monitor and analyze their free throw performance through detailed metrics.
- Utilized the **Python** and the **nba_api** for web scraping NBA player's data, creating a comparative analysis tool that aligns user shooting data with professional NBA players free throw performance.

EDUCATION

University of Colorado, Boulder | Boulder, CO

Expected Graduation May 2028

M.S. Computer Science

University of Wisconsin, Madison | Madison, WI

Graduation Dec 2023

B.S. Double Major in Computer Science and Data Science

SKILLS & TECHNICAL TOOLS

Languages: Python, SQL, JavaScript, Java, HTML/CSS, R, Bash

Tools: Pandas, Numpy, Dash, Streamlit, ReactJS, React Native, Flask, Tableau, Figma, Excel, Matplotlib, Seaborn, Plotly

Developer Tools: Git, Github, Jupyter Notebook, Visual Studio Code, Vercel, Postman, Swagger

Databases: Snowflake, Firebase, MySQL, SQLite